

AFCMiner: Finding Absolute Fair Cliques From Attributed Social Networks for Responsible Computational Social Systems

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Abstract—Cohesive subgraph mining on attributed social networks is attracting much attention in the realm of graph mining and analysis. Most existing studies on cohesive subgraph mining over attributed social networks neglect the fairness of attributes, which lead to difficulties in deploying responsible applications. Toward this end, this article formulates a new problem by introducing fairness into cliques model to mine the absolute fair cliques from attributed social networks. Specifically, this article adopts formal concept analysis (FCA) methodology to represent the given attributed social network, and extracts a set of special attributed equiconcepts to further return the absolute fair maximal cliques. Then, we develop an efficient absolute fair cliques detection algorithm *AFCMiner* for the cases of single-dimensional attributed social networks, multivalued attributed social networks, as well as multidimensional attributed social networks. Extensive experiments are conducted for demonstrating that the proposed *AFCMiner* algorithm can significantly reduce the time for finding absolute fair cliques with the correctness guarantee. Finally, a case study is also presented for uncovering the usefulness of our model.

Index Terms—Absolute fair clique, attributed social network, fairness, formal concept analysis (FCA), responsible AI.

I. INTRODUCTION

BIG data era has witnessed the emergence of heterogeneous data including structured data, semi-structured data, and nonstructured data that are generated from both IoT systems and cyber-physic-social systems [1], [2]. Particularly, the graph-structured data that naturally represent the connections among objects, are widely used in our daily lives. For

instance, the Facebook and Sina Weibo provide an efficient communication platform and form an online social network between social media users; thousands of pages on the Internet constitute the WWW; transportation between national cities forms a logistics network. To understand the topological structure of those various networks and extract valuable information from them, graph models and analysis techniques [3], [4] have been utilized for understanding the features and further conducting the potential applications of networks. Cohesive subgraph mining in networks, as a fundamental problem in graph mining and analysis, has attracted much attention from both academia and industry and achieves many promising application fruits, such as overlapping communities detection over social networks [5], protein complexes identification [6], and team formation [7].

Recently, the concept of fairness is widely concerned and has been incorporated into machine learning for advancing the responsible AI and responsible computational social systems [8], [9]. The existing research reported that the machine learning or responsible AI without considerations of fairness will lead to a series of bias and prejudice to decision-making and prediction. For instance, the video interview platform HireVue claimed that the possibility of the success of candidates in the work could be predicted, which was criticized in 2019. Electronic Privacy Information Center (EPIC) submitted a complaint against HireVue to the fair-trade commission, saying that the AI-driven candidate was “prejudiced, unable to prove, and cannot be copied.”¹ This prejudice and discrimination in recruitment and workplace is obviously illegal. In addition, an “elimination mosaic” (generating clear image) algorithm is applied into a low-resolution photograph of the first black President Barack Obama, and the output result of this AI algorithm is a white man [10]. Clearly, this kind AI algorithm may lead to the racial bias that is not allowed in various responsible AI-empowered systems.

Inspired by the conceptual idea of fairness in machine learning and AI, recent research works attempted to incorporate fairness factor to the field of fair graph mining [11], [12], [13], [14]. In fact, fair graph mining also has a number of potential responsible AI-empowered computational social applications [15], [16], [17], [18]. For example, a certain company plans to recruit a group of candidates from an

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¹<https://epic.org/documents/in-re-hirevue/>

attributed social network for completing a given interdisciplinary project effectively with the joint considerations of gender, educational background, as well as working field balance. In the extreme case, an optimal team with absolute fairness for this project should contain the same number of male and female candidates, the same portion of candidates with diverse educational background, and the same number of candidates with various working fields. From the fairness point of view, this kind of team formation can promote the collaboration and union among candidates for the effective project completion. Therefore, how to find this optimal team of candidates is becoming a challenging issue in attributed social networks analysis.

Technically, the above problem can be addressed by identifying the fair subgraphs from attributed social networks. However, the hardness of addressing this problem is NP-hard. Besides, multivalued or multidimensional attributes also lead to great challenges for identifying the absolute fair cliques from attributed social networks. In this work, we focus on the problem of finding absolute fair maximal cliques in attributed social networks where each vertex has one/multiple attributes with binary/multivalued. Concretely, we first formulate the problem on absolute fair maximal cliques detection in attributed social networks. Then, the detection methodologies and algorithms are developed for the cases of single-dimensional binary-valued attributed social networks, multivalued attributed social networks, as well as multidimensional attributed social networks. For revealing the practical usage of the problem addressed in this research, a case study is also conducted. Below, we summarize the major contributions of this research.

- 1) *Absolute Fair Clique Model*: By considering the various attributes of nodes in the attributed social network, we propose an absolute fair clique model to characterize the fairness of a cohesive subgraph in an attributed social network. Specifically, a distance-aware measure for evaluating the fairness of cliques is given. Then, we formulate the absolute fair cliques enumeration problem and mathematically prove that it is a NP-hard problem.
- 2) *Formal Concept Analysis (FCA)-Based Absolute Fair Clique Detection Approach*: We first define a new concept called attributed equiconcept which characterizes the topological structure and attributes information of a cohesive subgraph. Since equiconcepts are exactly matching with maximal cliques in the graph, thus attributed equiconcept can provide a good clue for enumerating the absolute fair cliques from attributed social networks. With this clue, a theorem and a corresponding algorithm for obtaining absolute fair maximal cliques based on FCA from an attributed social network are presented.
- 3) *Evaluation*: We conduct extensive experiments including an illustrative example on a real-world dataset of job-seeking social application. To be specific, the trend of processing time and size of absolute fair cliques are analyzed by adjusting the subset sizes. Concretely, the number of absolute fair cliques increases with the

increase of graph size. For the same attribute, the larger the attribute granularity is, the less absolute fair cliques are mined. For the different attributes, the more attributes being considered simultaneously, i.e., the higher the dimension of the attribute, the less absolute fair cliques are mined. In addition, for the nodes in the set of absolute fair cliques which we mined by *AFCMiner*, we provide a concept of “overlap” D_{overlap} to evaluate the importance of the nodes, the higher D_{overlap} of the nodes, the more important their roles in the network structure and vice versa.

The rest of this article is structured as follows. Section II revisits the state-of-the-art work on attributed graph mining and fairness-aware data mining. The fundamental definitions related to this work as well as the problem statement on absolute fair cliques detection in an attributed social network are presented in Section III. Section IV provides an FCA-based absolute fair cliques detection approach and the corresponding algorithms. The experimental results and analysis, and case study are demonstrated in Section V. Finally, Section VI concludes this article including our future work.

II. RELATED WORK

Since our research falls within the scope of attributed graph mining and fairness-aware data mining, this section will overview the state-of-the-art in these two directions.

A. Attributed Graph Mining

Attributed graph mining, as an important research direction in the field of graph mining, improves the complexity of mining algorithm and enlarges the range of applications by integrating attributes of nodes and edges [19]. The relevant research work includes attributed graph representation [20], access and query [21], and pattern recognition [22].

Attribute graph representation refers to encoding the topology and attribute information of nodes or edges in the graph into the low-dimensional vector space. Zhang et al. [23] utilized a neural network autoencoder with a dual optimization objective, which encodes both topology and attribute information as a single vector. In addition, the research [24] utilizes the random hash algorithm of node position (structure) and attribute information in the attributed graph, and represents the nodes using hash values.

The access of attributed graph involves the process of data storage, indexing and retrieval. In terms of access, an attributed graph-oriented online processing system is developed, which supports distributed access, transaction, and rollback of graphs [21]. Xie et al. [25] proposed two influential community search algorithms by taking both the influence and node attributes into consideration. Specifically, a novel scoring function for checking the cohesiveness of the community was defined. Then, their method could find the attributed pkd-truss community by maximizing the attribute and influence relevance scoring function.

Pattern recognition in attributed graphs refers to the detection of substructures that meet the requirements in attribute

graphs, including frequent pattern detection and community detection. Lin et al. [26] developed a novel multiview attributed graph clustering framework which exploits both node attributes and graphs. Li et al. [27] presented a new problem called tenuous attributed group query, and a new indicator, k -tenuity, to measure the structural tenuity of a group in attributed networks. Considering the fuzzy features of relations between nodes, our previous work [28] pioneered a novel problem of skyline (λ, k) -cliques identification from a fuzzy attributed social network. Then, a fuzzy FCA-based skyline (λ, k) -cliques identification algorithm is developed. In this article, we study the problem of finding tenuous groups in attributed networks that contain specific vertices.

B. Fairness-Aware Data Mining

Recent years have witnessed tremendous growth in fairness-aware machine learning and its applications [29], [30], [31], [32]. Fair machine learning aims to mitigate the biases of model predictions against certain subpopulations regarding sensitive attributes such as race and gender. Diana et al. [33] developed a provably convergent, practical algorithm for addressing minimax group fairness problems and error minimization problems subject to group-dependent error upper bound constraints. Cheng et al. [34] studied fairness in committee selection problems by considering fairness via group stability. Ma et al. [35] proposed a new fairness notion-graph counterfactual fairness, which considers the potential biases brought by different causal pathways from sensitive attributes to the graph model predictions. Then, a novel framework GEAR was developed to learn node representations toward graph counterfactual fairness. Pan et al. [13] first incorporated the fairness into graph mining and proposed a weak fair clique and a strong fair clique model to characterize the fairness of a cohesive subgraph.

Different from the previous attributed graph models, this article pioneers a promising absolute fair clique model, which characterizes both structural property and fairness of nodes simultaneously. Aiming to enumerate the absolute fair cliques from an attributed social network, FCA methodology is adopted for extracting attributed equiconcepts and further filtering the resulting absolute fair cliques by taking both fairness and maximality into account.

III. PROBLEM STATEMENT

This section first revisits the fundamental definitions utilized in this research. Then, the problem of absolute fair cliques enumeration in an attributed social network is mathematically formulated.

Definition 1 (Attributed Social Network): An attributed social network can be represented as $G = (V, E, A)$ with V indicating the set of vertices, i.e., $V = \{v_1, v_2, \dots, v_m\}$, E indicating the set of edges $E = \{e_{ij}\}$, for each vertex $v \in V$, it has an attribute A with its value $v \cdot \text{val}$. Here, A_{val} denotes the set of attributes on vertices, $A_{\text{val}} = \{v \cdot \text{val} | v \in V\}$.

Definition 2 (Clique): Given an attributed social network G , a clique C of G is a subgraph where each pair of vertices

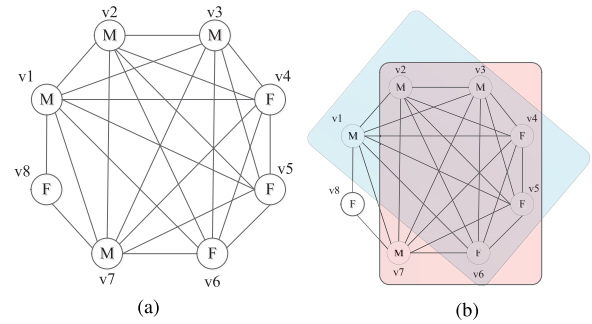


Fig. 1. Attributed social network and its fair cliques. (a) G . (b) Fair cliques in G .

are connected. Formally, for any two vertices $v_i \in C$ and $v_j \in C$, there exists an edge e_{ij} between them.

Definition 3 (Absolute Fair Clique): Given an attributed social network G , a clique C of G is an absolute fair maximal clique if the number of vertices for each $a_i \in A_{\text{val}}$ is exactly the same.

Definition 4 (Absolute Fair Maximal Clique): Given an attributed social network G , a clique C of G is an absolute fair maximal clique if it satisfies: 1) **(Fairness)** the number of vertices for each $a_i \in A_{\text{val}}$ is exactly the same and 2) **(Maximal)** there is no clique $C' \supset C$ satisfying (1).

Example 1: Let us consider a research collaboration network including male and female scientists. Formally, it can be represented as an attributed social network $G = \{V, E, A\}$ with $A_{\text{val}} = \{M, F\}$ in Fig. 1(a). According to Definition 3, we know that subgraphs $C_1 = \{v_1, v_2, \dots, v_6\}$ and $C_2 = \{v_2, v_3, \dots, v_7\}$ are the fair cliques since the number of vertices with attribute “M” is 3, and with attribute “F” is also 3. Note that they are also absolute fair maximal cliques since they satisfy there is no subgraph C' that contains C as defined in Definition 4. In other words, absolute fair cliques consider the gender equality and guarantee the same number of male and female scientists in research collaboration networks.

Problem Statement: Given an attributed social network G , the goal of this work is to enumerate all absolute fair cliques and absolute fair maximal cliques, respectively.

Example 2: Continue Example 1, we can easily know that the subgraphs $\{v_1, v_2, v_4, v_5\}$, $\{v_2, v_3, v_4, v_5\}$ are both absolute fair cliques by Definition 3. However, there exist four maximum absolute fair cliques in G , i.e., $C_1 = \{v_1, v_2, \dots, v_6\}$, $C_2 = \{v_2, v_3, \dots, v_7\}$, $C_3 = \{v_1, v_2, v_4, v_5, v_6, v_7\}$, and $C_4 = \{v_1, v_3, v_4, v_5, v_6, v_7\}$. Obviously, all maximum absolute fair cliques are the special forms of absolute fair cliques.

Theorem 1: The hardness of problem on finding all absolute fair (maximal) cliques is NP-hard.

Proof: The problem on finding all absolute fair cliques can be viewed as a special case of cliques enumeration which is NP-hard [36]. Thus, enumerating all absolute fair cliques is also an NP-hard problem. Besides, absolute fair maximal cliques is a subset of absolute fair cliques since they have to meet the maximality. Therefore, absolute fair maximal cliques enumeration is more challenging than enumerating absolute fair cliques. Based on the above analysis, we can easily prove that the hardness of problem on finding all absolute fair (maximal) cliques is NP-hard. ■

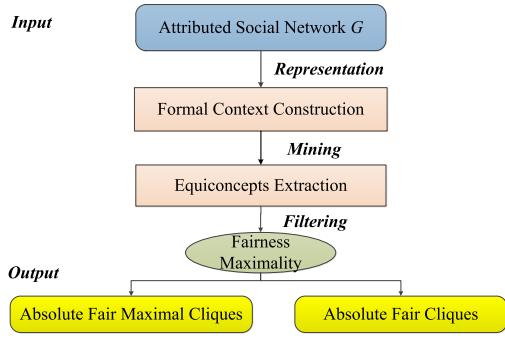


Fig. 2. Framework of FCA-based absolute fair (maximal) cliques enumeration from an attributed social network.

IV. ALGORITHM FOR ABSOLUTE FAIR CLIQUES ENUMERATION

In this section, we focus on presenting algorithms for enumerating absolute fair cliques given the input attributed social network G .

A. FCA-Based Absolute Fair Cliques Enumeration

Due to the powerful modeling ability of FCA for social network analysis, this section will provide implementation approach and algorithm of FCA-based absolute fair cliques enumeration. First, a framework of FCA-based absolute fair cliques enumeration from an attributed social network is presented. Then, the developed algorithm and its time complexity are proposed and further analyzed.

1) *Framework of FCA-Based Absolute Fair (Maximal) Cliques Enumeration From an Attributed Social Network:* As depicted in Fig. 2, the framework of FCA-based Absolute Fair (maximal) Cliques Enumeration from an Attributed Social Network takes the attributed social network G as the input, then it constructs the formal context of G by our previous representation approach [37]. With the constructed formal context of G , the corresponding concept lattice is built; and the equiconcepts which are exactly matching with maximal cliques are then extracted; By taking the fairness and maximality into account, the absolute fair cliques and maximal cliques can be obtained.

2) *Formal Context Construction:* Recent years have witnessed the success on social network analysis with FCA methodology. To make FCA methodology applicable to modeling an attributed social network, the nodes are regarded as objects, and both nodes and their attributes are regarded as attributes in the constructed formal context. Hence, a formal context of an attributed social network can be formalized as $K = (V, V \cup A, R)$ by the following incidence matrix, in which R contains the topological relationships among nodes as well as the attribute relationships between V and A .

Definition 5 (Incidence Matrix): Let G be an attributed social graph with n vertices that are assumed to be ordered from v_1 to v_n . The $n \times (n + k)$ matrix M is an incidence matrix, in which

$$M = \begin{cases} m_{ij} = 1, & \text{if } (v_i, v_j) \in E, \text{ or } (v_i, v_{\text{val}}) \in R \\ m_{ij} = 1, & \text{if } i = j, \text{ or } (v_i, v_{\text{val}}) \in R \\ m_{ij} = 0, & \text{otherwise} \end{cases} \quad (1)$$

TABLE I

$V/V \cup A$	v_1	v_2	v_3	v_4	v_5	v_6	v_7	v_8	M	F
v_1	1	1	1	1	1	1	1	1	1	0
v_2	1	1	1	1	1	1	1	0	1	0
v_3	1	1	1	1	1	1	1	0	1	0
v_4	1	1	1	1	1	1	1	0	0	1
v_5	1	1	1	1	1	1	1	0	0	1
v_6	1	1	1	1	1	1	1	0	0	1
v_7	1	1	1	1	1	1	1	1	1	0
v_8	1	0	0	0	0	0	1	1	1	0

where n is the number of vertices in G , i.e., $|V| = n$, k denotes the number of attribute values for vertices, i.e., $|A \cdot \text{val}| = k$, and $(v_i, v_{\text{val}}) \in I$ indicates that the vertex v_i has the attribute v_{val} .

Example 3: Reconsider the attributed social network G in Fig. 1(a). By Definition 5, the formal context of G can be constructed as shown in Table I.

The above matrix M is composed of two sub matrices M_{adj} and M_{att} : the matrix M_{adj} is used to characterize the topological relationships and another matrix M_{att} is used to describe the attribute relation of vertices. Apparently, the matrix M_{adj} is a symmetry matrix with all diagonal elements marked with “1.”

3) *Attributed Equiconcepts Extraction:* Inspired by our previous results [38] that the equiconcepts are exactly matching with the maximal cliques in social networks, this section focuses on defining and extracting attributed equiconcepts, and presenting the equivalence between attributed equiconcepts and absolute fair maximal cliques.

Definition 6 (Attributed Concepts Derivation Operators): For a formal context of an attributed social network $K = (V, V \cup A, R)$, the attributed concepts derivation operators $\uparrow$ and $\downarrow$ on $X \subseteq V$ and $B \subseteq V$ are defined as follows:

$$X^\uparrow = \{u \in V \mid \forall u \in X, (u, u) \in M_{\text{adj}}\} \quad (2)$$

$$B^\downarrow = \{x \in V \mid \forall a \in B, (x, a) \in M_{\text{att}}\}. \quad (3)$$

With the above-defined attributed concepts derivation operators, the attributed equiconcept can be defined as follows.

Definition 7 (Attributed EquiConcept): For a formal context of an attributed social network $K = (V, V \cup A, R)$, if a triple (X_1, X_2, B) satisfies $X_1^\uparrow = X_2$, $X_2^\uparrow = X_1$, $X_1 = X_2$, and $X_1^\downarrow = B$, then (X_1, X_2, B) is called as an attributed equiconcept, here X_1 , X_2 , and B denote the extent, intent, and attribute information of this attributed equiconcept, respectively.

Definition 8 (Attributed K -EquiConcept): For a formal context of an attributed social network $K = (V, V \cup A, R)$, if a triple (X_1, X_2, B) satisfies $X_1^\uparrow = X_2$, $X_2^\uparrow = X_1$, $X_1 = X_2$, $|X_1| = X_2 = K$, and $X_1^\downarrow = B$, then (X_1, X_2, B) is called as an attributed K -equiconcept, here X_1 , X_2 , and B represent the extent, intent, and attribute information of this attributed equiconcept, respectively.

Example 4: Based on the constructed formal context as shown in Table I, we construct the following concept lattice for an attributed social network G .

Clearly, there are attributed concepts, i.e., $(\{1, 2, 3, 4, 5, 6, 7, 8\}, \{1, 7\}, \emptyset)$, $(\{1, 7, 8\}, \{1, 7, 8\}, M)$, $(\{1, 2, 3, 4, 5,$

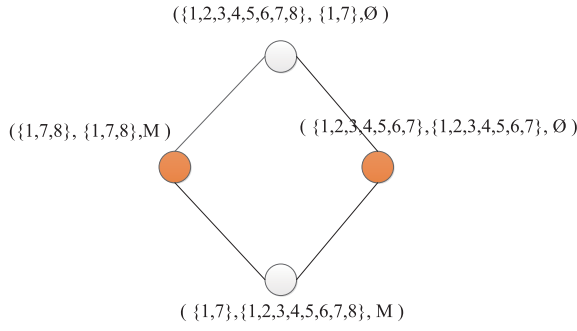


Fig. 3. Concept lattice of an attributed social network.

6, 7}, {1, 2, 3, 4, 5, 6, 7}, ∅), and ({1, 7}, {1, 2, 3, 4, 5, 6, 7, 8}, M). Note that the attributed concepts ({1, 7, 8}, {1, 7, 8}, M) and ({1, 2, 3, 4, 5, 6, 7}, {1, 2, 3, 4, 5, 6, 7}, ∅) are the attributed equiconcepts.

4) *Filtering Absolute Fair Cliques From Attributed Equiconcepts*: Our previous research has proved that maximal cliques can be derived from its equiconcepts [38], this section intends to extend it to assist us identifying absolute fair maximal cliques from attributed equiconcepts. Importantly, a theorem for obtaining absolute fair maximal cliques from an attributed social network is presented.

Theorem 2: Given an attributed social network $G = (V, E, A)$, the absolute fair maximal cliques can be obtained based on attributed equiconcepts (X_1, X_2, B) under the following cases.

- 1) If $|X_1|(|X_2|) \bmod |A_{\text{val}}| = 0$ and the number of vertices for each $a_i \in A_{\text{val}}$ is exactly the same, then X_1 (X_2) is the absolute fair maximal clique.
- 2) If $|X_1|(|X_2|) \bmod |A_{\text{val}}| \neq 0$, then X_1 (X_2) is not an absolute fair maximal clique.

Proof: As we known, the attributed equiconcept (X_1, X_2, B) is corresponding to the maximal clique. To guarantee the fairness for further extracting absolute fair maximal cliques, we should ensure that the number of vertices for each $a_i \in A_{\text{val}}$ is exactly the same. Let us suppose the vertices have $|A_{\text{val}}|$ attributes, thus $|X_1|$ or $|X_2|$ should satisfy $|X_1|/|X_2| \bmod |A_{\text{val}}| = 0$. For example, if $|A_{\text{val}}| = 3$, then the attributed equiconcept $(\{1, 7, 8\}, \{1, 7, 8\}, M)$ in Fig. 3 could be the candidate of absolute fair maximal clique. On the contrary, the attributed equiconcept $(\{1, 2, 3, 4, 5, 6, 7\}, \{1, 2, 3, 4, 5, 6, 7\}, \emptyset)$ in Fig. 3 must not the candidate of absolute fair maximal clique since it does not meet $|X_1|/|X_2| \bmod |A_{\text{val}}| = 0$. ■

Lemma 1: Given an attributed social network $G = (V, E, A)$, the absolute fair cliques can be derived from attributed equiconcepts (X_1, X_2, B) . That is to say, the absolute fair cliques are hidden in the power set of X_1 , $P(X_1)$.

Example 5: Let us continue Example 4, we know that $|A_{\text{val}}| = |M, F| = 2$, the attributed equiconcepts $(\{1, 7, 8\}, \{1, 7, 8\}, M)$ and $(\{1, 2, 3, 4, 5, 6, 7\}, \{1, 2, 3, 4, 5, 6, 7\}, \emptyset)$. Since they don't satisfy $|X_1|/|X_2| \bmod |A_{\text{val}}| = 0$, they are not absolute fair maximal cliques according to Theorem 2. That is to say, there is no absolute fair maximal cliques in this example. However, absolute fair cliques might be derived from the power

Algorithm 1 *AFCMiner*

Input: An attributed social network $G = (V, V \cup A, R)$

Output: A set of Absolute Fair cliques ζ

```

1 Initialize  $\zeta = \emptyset$ 
2 begin
3 Formal context  $K(G)$  construction via Incidence
  matrix  $M$ 
4 Build Attributed concept lattice  $L(K(G))$ 
5 end
6 for each attributed concept  $(X_1, X_2, B) \in L(K(G))$ 
7 begin
8   if  $X_1 = X_2$  and  $|X_1| = |X_2|$  then
9     if FairnessFilter $((X_1, X_2, B))$  then
10        $\zeta \leftarrow \zeta \cup (X_1, X_2, B)$ 
11   else
12      $P(X_1) \leftarrow \text{Attributed Concepts Derivation}(X_1)$ 
13     if Fairness Filter $(P(X_1))$  then
14        $\zeta \leftarrow \zeta \cup P(X_1)$ 
15 end
```

set of attributed equiconcepts $(\{1, 7, 8\}, \{1, 7, 8\}, M)$ and $(\{1, 2, 3, 4, 5, 6, 7\}, \{1, 2, 3, 4, 5, 6, 7\}, \emptyset)$. For instance, we can easily obtain the following maximum absolute fair cliques $C_1 = \{v_1, v_2, \dots, v_6\}$, $C_2 = \{v_2, v_3, \dots, v_7\}$, $C_3 = \{v_1, v_2, v_4, v_5, v_6, v_7\}$, $C_4 = \{v_1, v_3, v_4, v_5, v_6, v_7\}$.

B. Algorithm Description

With the above proposed FCA-based absolute fair cliques enumeration approach, this section is devoted to presenting the algorithm for absolute fair cliques enumeration from an attributed social network. This section mainly presents 3 pseudo-code of our implementation procedures, including: 1) absolute fair cliques miner (*AFCMiner*); 2) fairness filter; and 3) attributed concepts derivation.

Algorithm 1 proceeds as: First, the inputs of the algorithm is an attributed social network G , then, a set of absolute fair cliques with ζ is initialized in line 1. After that, it goes into the procedures of formal context construction via Incidence Matrix (which is mentioned in Definition 5), and build an attributed concept lattice (Lines 3 and 4). In lines 6–14, iterate through all attribute concepts and determine whether they are attribute equiconcepts (i.e., the corresponding maximal cliques in the graph); if so, call Algorithm 2 to determine whether the clique satisfies Definition 3, and if not, call Algorithm 3 to derive sub-concepts and determine whether they are absolute fair clique (to satisfy Definition 3), finally, the absolute fair maximal cliques and absolute fair cliques are inserted into ζ .

Algorithm 2 is a subroutine of Algorithm 1 that determines whether the attributes of cliques in Algorithm 1 are fair. Its working process is described as follows. First, input a clique (or an Equiconcept), then check whether it is an attribute fair concept based on Theorem 2 (Line 1). Lines 2–6 traverse the attribute set A_{val} , counting the numbers of each attribute contained by this clique X_1 , and storing them in set *AiNumber*.

Algorithm 2 *Fairness Filter Algorithm*

Input: (X_1, X_2, B)
Output: *True* or *False*

```

1 if  $|X_1|(|X_2|) \bmod |A_{val}| = 0$ 
2   for each attribute  $a_i$  in  $A_{val}$ 
3     for each vertex  $v$  in  $X_1$ 
4       if  $B(v) = a_i$  then
5          $count_i \leftarrow count_i + 1$ 
6        $A_iNumber \leftarrow A_iNumber \cup count_i$ 
7  $p \leftarrow 0$ ,  $result \leftarrow True$ 
8 do  $Num \leftarrow A_iNumber[p]$ 
9   if  $Num \neq A_iNumber[p++]$  then
10      $result = False$ , break
11 while  $A_iNumber[p]$ 
12 return  $result$ ;
```

Algorithm 3 *Attributed Concepts Derivation Algorithm*

Input: X_1
Output: A set of $P(X_1)$

```

1  $P(X_1) \leftarrow \{\}$ 
2 for each vertex  $v_i$  in  $X_1$ 
3   for subset in  $P(X_1)$ 
4      $Temp \leftarrow subset \cup v_i$ 
5    $P(X_1) \leftarrow P(X_1) \cup Temp$ 
6 return  $P(X_1)$ ;
```

Lines 7–11 determine whether the number of each attribute is equal, returning *True* if so, otherwise return *False*.

According to Lemma 1, the absolute fair cliques are hidden in the power set of X_1 , so the purpose of Algorithm 3 is to calculate and return the power set of X_1 , $P(X_1)$.

C. Time Complexity Analysis

In this section, we discuss the time complexity of *AFCMiner* in attributed social networks, which identify the absolute fair cliques. The time complexity of Algorithm 1 for identifying the absolute fair cliques is composed of the time complexity of maximal cliques detection by FCA method, and the time complexity of Algorithms 2 and 3. First, the time complexity of maximal cliques detection by FCA is $\Theta(|V|^3)$, $|V|$ is the number of vertices of Graph G . We denote γ as the number of maximal cliques, m as the number of attribute types contained in attribute matrix $|M_{att}|$, k as the number of nodes in *Attributed K-Equiconcept*. Hence, the time complexity of Algorithm 2 is $\Theta(m \times k \times |\gamma|)$. In the worst case, the Algorithm 3 traverses all maximal cliques and generate the power set of all maximal cliques, the time complexity of Algorithm 3 is $\Theta(k^2 \times |\gamma|)$. In summary, the time complexity of Algorithm 1 is $\Theta(|V|^3 + m \times k \times |\gamma| + k^2 \times |\gamma|)$.

D. Baseline Model

In this section, we apply the Bron–Kerbosch algorithm (BK) as the *BaseLine* algorithm to detect the maximal cliques in a social network. BK method is a famous and classical recursive

Algorithm 4 *BK Algorithm for Mining Absolute Fair Cliques*

Input: graph $G = (V, E)$
Output: The absolute fair cliques set ζ

```

1 Initial sets  $R, X$  as empty sets, set  $P = \{v\}$ ,  $v \in V$ 
2 begin Bron-Kerbosch( $R, P, X$ ):
3   if  $P$  and  $X$  are both empty:
4     report  $R$  as a maximal clique
5      $maximalcliques \leftarrow R$ 
6   for each vertex  $v$  in  $P$ :
7     Bron-Kerbosch( $R \cup \{v\}$ ,  $P \cap N(v)$ ,  $X \cap N(v)$ )
8      $P \leftarrow P \setminus \{v\}$ 
9      $X \leftarrow X \cup \{v\}$ 
10 end
11 begin
12 for each maximal clique  $c$  in  $maximalcliques$ 
13   if FairnessFilter( $c$ ) then
14      $\zeta \leftarrow \zeta \cup c$ 
15 else
16    $DerivedClique \leftarrow Attributed\ Concepts\ Derivation$ 
17     ( $c$ )
18   if Fairness Filter( $DerivedClique$ ) then
19      $\zeta \leftarrow \zeta \cup DerivedClique$ 
19 end
```

backtracking algorithm designed by Dutch scientists Bron and Kerbosch [39], the aim is to enumerate all maximal cliques in a graph G . The main idea of BK method is giving three disjoint sets of vertices R, P, X at first, find a maximal clique containing all vertices in R , some vertices in P , and vertices not in X . At each iteration of the algorithm, set P and X are disjoint sets, and their union consists of those vertices that form cliques when added to R . Before starting the recursive algorithm, set R and X to the empty set, and set P to the vertex set of the graph. In each recursive call, the algorithm considers the nodes in P in turn. If there is no such vertex v , there are two cases: 1) if X is empty, report R as the maximal clique and 2) if X is not empty, backtrack. For each node v selected from P , it will make a recursive call, where v is added to R , meanwhile, P and X are restricted to v 's neighbor set $N(v)$, which will find and report that R contains all clique expansions of v . Then, moving v from P to X to exclude it in later cliques and continues with the next vertex in P .

In the pseudo-code of *Bron–Kerbosch* algorithm, at first, we initialize the sets R and X as empty sets, and initialize set P to store all vertices in the graph (Line 1). In the lines 3 and 4, if both P and X are empty, then R is the maximal clique. Iterate over each vertex in the set P , merge $\{v\}$ with R , take the intersection of P and $N(v)$, and the intersection of X and $N(v)$ in lines 5 and 6, where $N(v)$ denotes the neighbor nodes of the current node in this iteration. Then, backtracking the process after finding v . In line 7, after the operation on v is finished, delete v from the unprocessed point set P , and add the vertex v to the X set in lines 8 and 9. In lines 12–18, set the maximal cliques as an extent of concept, which are generated by Bron–Kerbosch, then we utilize our

Attributed Concepts Derivation Algorithm to derive candidate cliques and filter absolute fair cliques.

V. EXPERIMENTS AND ANALYSIS

This section is devoted to evaluating the proposed method using real-life datasets and conducting an illustrative example to demonstrate the utility of our method. The experiment aims to verify the effectiveness of the proposed *AFCMiner* algorithm, so that we apply *AFCMiner* on an attributed social network, which is formed by a large job-seeking social application in China.

A. Experiment Setup

In our work, all algorithms are implemented with Python programming language and are run on an Intel² Core³ i7-10700 at 2.90 GHz, 32 GB RAM computer.

Initially, we clarify the composition of the datasets of the China's largest online professional job-seeking social network [40], which is described as an attribute social network, including the users' friendship dataset, the educational background, the work field, and the gender information of users.

To verify the practicability of our method in both two-valued and multivalued (i.e., the number of attribute values is greater than 2) attribute scenarios, we experiment with these datasets to test performances. Among them, gender is a two-valued attribute, which is categorized as male and female; Educational background is also a two-valued attribute, it has two categories, 985-211 university and others (a category method of Chinese universities, which represents the educational level of the school. In general, 985-211 universities have higher teaching quality); Work field is a multivalued attribute, including five categories: information technology (IT), Finance, Media, and human resources management (HRM).

Especially, since gender is a two-valued attribute (Male/Female), we calculate the absolute fair cliques with gender in Experiment-1 to evaluate the correctness and efficiency of our method with the two-valued attribute; In a similar way, since the work field is a multivalued attribute that can be divided into two-five valued attributes according to different categorization methods, we apply it to Experiment-2 to verify the correctness and efficiency validation of the absolute fair cliques of our method for multivalued attributes.

Further, we consider the fairness of multidimensional attributes (i.e., simultaneously ensuring the fairness of attributes such as gender, work field, and educational background of a clique) by ensuring the correctness and efficiency of *AFCMiner* in multivalued attribute datasets. It is shown as Experiment-3.

1) *Dataset I*: This is a user social network of a job-seeking social application, which we call the user friendship dataset. It contains the users' friendship information of the job-seeking social application (we regard a user's following or being followed as a friendship of this user), in the social network, each node represents a user, and the edge between

TABLE II
DATASETS I. FRIENDSHIP OF USERS

Sub-Dataset	Node	Edge	d_{avg}	d_{max}
SubSet 1	250	644	5.15	50
SubSet 2	500	2387	9.55	101
SubSet 3	750	3977	10.60	140
SubSet 4	1000	6131	12.26	184
SubSet 5	1250	8757	14.01	216
SubSet 6	1500	12209	16.28	261

TABLE III
DATASETS II. CATEGORIES OF DIFFERENT ATTRIBUTES

DataSets	Category	Tag	Number
Gender	Male	1	1261
	Female	2	239
Education	985-211	1	657
	Others	2	843
Work Field	IT	1	1250
	Finance	2	58
	Media	3	110
	HRM	4	12
	Others	0	70

the two nodes represents their friendship. Since our work is devoted to the study of undirected graphs, we do not consider the direction of the network edge. The network contains 1500 nodes and 12 209 edges. In our experiment, to test the time performance of our algorithm and analyze the method performance under different scales of datasets, we divided the 1500 nodes into six sub-datasets, as shown in Table II below.

2) *Dataset II*: The dataset contains the attributes information of users, the categories are shown in Table III. It can be seen from Table III that the distribution of the categories of attributes is uneven. For example, the ratio of male to female of the gender attribute is close to 5.2:1.

3) *Dataset III*: This dataset describes the educational background of 1500 users in Dataset I. We divide the educational background of users into two categories: 985-211 universities and other universities. The specific categories of Dataset III are shown as Table III.

4) *Dataset IV*: This dataset contains the work fields of the users who are in Dataset I, which is categorized into five categories: IT, Finance, Media, HRM, and others. The specific categories of this dataset are shown in Table III.

B. Comparison System

As aforementioned, we provided BK method as a baseline method to detect absolute fair cliques. The main idea is to utilize the BK algorithm to enumerate all maximal cliques first, and then call Algorithms 2 and 3 to filter the fairness of the maximal cliques and derive absolute fair cliques, so as to obtain all absolute fair cliques. To evaluate the correctness and efficiency of our algorithm more comprehensively, in the comparison experiment, in addition to the BK algorithm, we adopt several other cliques detection/enumeration algorithms, and combined Algorithms 2 and 3 as the baseline algorithms. The maximal cliques/cliques enumeration algorithm descriptions are as follows.

²Registered trademark.

³Trademarked.

TABLE IV
EXPERIMENT-1. DIFFERENT SCALES OF NODES WITH GENDER

Dataset	# of Absolute Fair Maximal Cliques		# of Absolute Fair Cliques		# of All Absolute Fair Cliques
	BaseLine Methods	AFCMiner	BaseLine Methods	AFCMiner	
SubSet 1	59	59	122	122	181
SubSet 2	176	176	917	917	1093
SubSet 3	331	331	1717	1717	2048
SubSet 4	411	411	2655	2655	3066
SubSet 5	553	553	3827	3827	4380
SubSet 6	781	781	6862	6862	7643

TABLE V
EXPERIMENT-1. RUNNING TIME COMPARISON

Dataset	Running Time (Unit: ms)				
	BK	BK+	CE	IGraph	AFCMiner
SubSet 1	4972	42	69	335	10
SubSet 2	5348	470	1833	575	256
SubSet 3	7549	1957	3564	1236	256
SubSet 4	9645	3527	10332	3810	2177
SubSet 5	11440	6851	31346	8100	5520
SubSet 6	24010	19469	100495	20348	17462

- 1) *BK*: The BK algorithm is a recursive backtracking algorithm for searching all maximal cliques of a graph G .
- 2) *BK+*: The BK+ algorithm is an optimization of BK method, and it is implemented based on BK method, by combining with the adaptation and discussion by Tomita et al. [41] and Cazals and Karande [42].
- 3) *CE*: This is an all cliques enumeration method, which is implemented by Zhang et al. [43]. According to this algorithm, we will obtain all cliques, and continue to filter the absolute fairness cliques by Algorithm 2. Since it includes all cliques in the graph G , it will be unnecessary to derive cliques from the maximal cliques, so Algorithm 3 will not be called in the process.
- 4) *IGraph*: To provide another comparison experiment with all cliques enumeration method, we utilize the algorithm for enumerating all cliques in igraph's Python-based module [44] as a baseline algorithm.

C. Results Analysis

1) *Experiment-1 (Efficiency and Correctness Evaluation for Binary-Valued Attribute)*: In this part, we evaluate the running time and the correctness of the results by applying the *AFCMiner* and four *BaseLine* algorithms to different sizes of datasets with Tables II and III.

First, with the *BaseLine* and *AFCMiner* algorithms, we detect the number of absolute fair maximal cliques and absolute fair cliques for subsets 1–6 containing 250, 500, 750, 1000, 1250, and 1500 vertices, respectively, as shown in Table IV (# in the Table denotes the number), and our algorithm exactly matches the number detected by the algorithms in *BaseLine*, thus proving the correctness of our algorithm. Meanwhile, since our dataset is incremental, it is obvious to observe through Fig. 4 that the number of detected absolute fair cliques increases as the scale of the dataset increases. In Fig. 4, the blue, green, and orange bars represent the number of the absolute fair maximal cliques, the derived

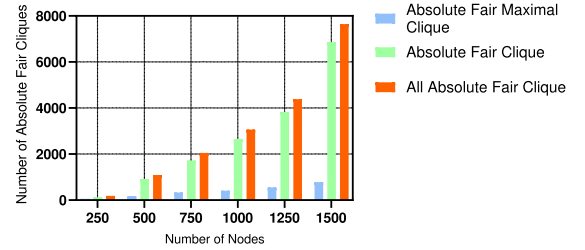


Fig. 4. Number of absolute fair maximal cliques with gender.

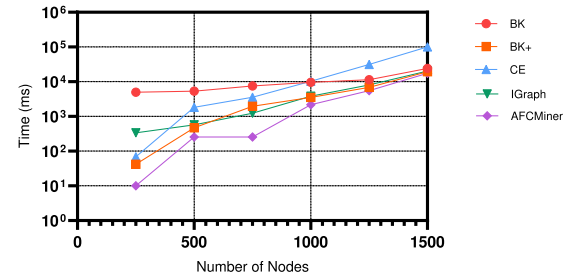


Fig. 5. Running time of *BaseLine* and *AFCMiner* method with gender.

absolute fair cliques, and all absolute fair cliques, respectively. It can be founded that although the number of absolute fair maximal cliques indicated by the blue bars increases with the number of nodes, the increase is not significant compared to the derived absolute fair cliques indicated by the green bars.

Then, we compare the running time of the *BaseLine* algorithms with the *AFCMiner* algorithm, and the results are shown in Table V, from which we see that the running time of our *AFCMiner* algorithm is significantly shorter than *BaseLine* when all the absolute fair cliques are detected. The visualized running time comparison results are shown in Fig. 5. The running time of detecting all absolute fair cliques is depending on how complex the structure of the attributed social network is (how many vertices and edges are in an attributed social network), For instance, the longest time is spent when the largest Subset-6 is applied.

2) *Experiment-2 (Efficiency and Correctness Evaluation for Multivalued Attribute)*: To evaluate the performance of the *AFCMiner* method under multivalued of the same attribute in the 1500 users' social network, meanwhile, to study the change in the number of absolute fair cliques at different granularity (numbers of category types), in this section, we empirically reclassify the work field dataset into various multivalued attributes, i.e., 2, 3, 4, and 5 V categories, the data are shown in Table VI. Among them, in the 2 V dataset, we label IT as 1 and others as 0; In the 3 V dataset, we label IT as 1, Finance, Media, and HRM as 2, and others as label 0; In the 4 V

TABLE VI
WORK FIELD DATASET

DataSets	Category	Tag	Number
2V	IT	1	1250
	Others	0	250
3V	IT	1	1250
	Finance&Media&HRM	2	180
	Others	0	70
4V	IT	1	1250
	Finance	2	58
	Media	3	110
	HRM	4	12
5V	IT	1	1250
	Finance	2	58
	Media	3	110
	HRM	4	12
	Others	0	70

TABLE VII

EXPERIMENT-2. NUMBER OF ALL ABSOLUTE FAIR CLIQUES IN DIFFERENT SCALES OF NODES WITH MULTIVALUED WORK FIELDS

Dataset	# of All Absolute Fair Cliques			
	2V	3V	4V	5V
SubSet 1	72	0	0	0
SubSet 2	308	0	0	0
SubSet 3	638	43	0	0
SubSet 4	1839	134	16	10
SubSet 5	3418	192	30	20
SubSet 6	6081	459	32	20

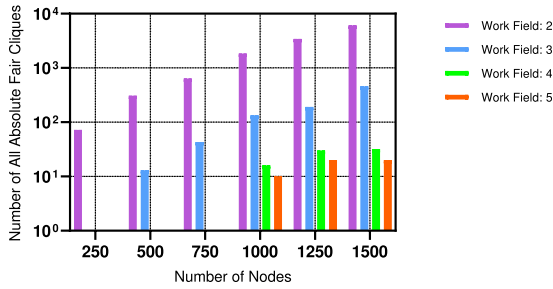


Fig. 6. Number of absolute fair maximal cliques with different level of multivalued work fields.

dataset, IT as 1, Finance as 2, Media as 3 and HRM as 4; The categories of 5 V remain the same as the one in dataset III.

We apply the user relationship social networks in subsets 1–6 of Table III and their corresponding 2, 3, 4, and 5 V datasets to our method to obtain the all absolute fair cliques of multivalued categories under the work field attributes. Table VII shows the results, the number of all absolute fair cliques in different scales subsets with 2, 3, 4 and 5 V work fields' attribute is represented in the following.

According to Table VII, we can obviously find that when the granularity of the attribute is larger (the more categories), fewer absolute fair cliques are detected. The visualized results are shown in Fig. 6, where the orange bar and green bar represent the number of all absolute fair cliques when the granularity of the work field attribute is 5 and 4, respectively, which is significantly lower than the blue bar with a granularity of 3 and the purple bar with a granularity of 2.

Meanwhile, we also compare the running time of the *BaseLine* algorithms with our algorithm in the working fields' data subsets 1–6 under the four granularities of 2–5 V. It is easily found by Fig. 7 that our algorithm (purple lines)

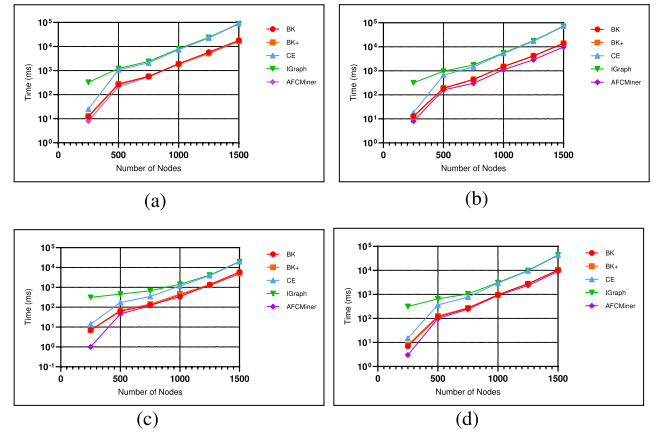


Fig. 7. Running time in multivalued work fields. (a) Two-valued. (b) Three-valued. (c) Four-valued. (d) Five-valued.

performs well in experiments at various granularities, and the running time is lower than the *BaseLine* algorithms.

3) *Experiment-3 (Absolute Fair Cliques Detection for Multidimensional Case)*: In addition to being applicable to the mining of absolute fair cliques of multivalued attributes, the method in this article also supports the mining of absolute fair maximal cliques of multidimensional attributes, that is, a maximal clique can guarantee the fairness of multiple attributes at the same time. Therefore, in Experiment-3, we consider the gender, work fields and educational background of 1500 users at the same time, intending to detect the cliques that can guarantee “fairness” in these 3-D attributes.

Based on above Experiment-2, The “work area” attribute has a variety of granularities (2–5 V), so a question arises as to how to choose the granularity of the attribute. Since gender and educational background are both two-valued attributes, and we prefer an even granularity for the work field categories, hence, we choose both 2 and 4 V granularities for comparison at first.

In Table VIII, we select work fields with granularity 2 and 4, respectively, to mine absolute fair cliques and also consider the “absolute fairness” of these cliques with educational background attributes. Obviously, when the granularity of the work fields is 4 (four-valued work fields), it is almost impossible to detect absolute fair cliques that satisfy both “fairness” of educational background and four-valued work fields. This phenomenon is highly potential in real life, due to the uneven distribution of attributes. However, in the section, to better observe experimentally the results of considering “fairness” by gender, educational background, and work fields simultaneously, we select the two-valued work fields' attribute.

Fig. 8 visualized the number of all absolute fair cliques mined based on single-dimensional attributes (educational background, gender, two-valued work fields) and multidimensional attributes (educational background & two-valued work fields & gender) in a social network of 1500 users. Among them, the blue bars indicate the number of absolute fair cliques of educational background, the purple bars indicate the number of absolute fairness cliques of gender, green bar is corresponding to two-valued work fields, and the orange bar indicates the number of multidimensional attributes. Fig. 8(a)–(c) are all cases of binary-dimensional attributes, and Fig. 8(d) is a case

TABLE VIII
EXPERIMENT-3. NUMBER OF ALL ABSOLUTE FAIR MAXIMAL CLIQUES WITH MULTIDIMENSIONAL CASE

Dataset	Education	Work Fields: 4-valued	Work Fields: 2-valued	Education & Work Fields: 4-valued	Education & Work Fields: 2-valued
SubSet 1	228	0	72	0	31
SubSet 2	228	0	308	0	31
SubSet 3	3244	0	638	0	283
SubSet 4	6696	16	1839	3	853
SubSet 5	12452	30	3418	3	1462
SubSet 6	26401	32	6081	3	2391

TABLE IX

OVERLAPPING DEGREE WITH EXAMPLE DATASETS

ID	532	565	754	1246	1533	1781	1834
$D_{overlap}$	0.33	0.33	1	0.66	1	0.33	0.33
Ranking	3	3	1	2	1	3	3

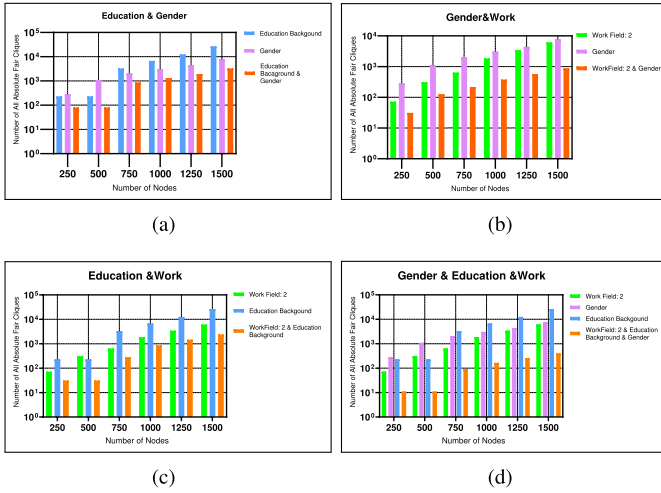


Fig. 8. Number of absolute fair maximal cliques with multidimensional attributes. (a) Education and Gender. (b) Gender and Work. (c) Education and Work. (d) Gender, Education, and Work.

of 3-D attributes. The orange bar in Fig. 8(d) is significantly lower compared to Fig. 8(a)–(c).

D. Case Study

To illustrate the practicality of *AFCMiner* in real life, a case study is given and conducted in this section. We use 2000 users of this job-seeking social application to assign an optimal team of candidates for a startup company, where it must be ensured that this team is fair in terms of gender, educational background, and work fields (four-valued: IT, Finance, Media, HRM), respectively. The dataset consists of 2000 nodes (represents the 2000 users) and 22 761 edges (represents the friendship between users), and 2000 records of gender, educational background, and work fields, corresponding to each of these 2000 nodes. The average degree d_{avg} is 22.761 and the maximum degree d_{max} is 357.

Then we apply *AFCMiner* to the dataset, where the number of absolute fair cliques for the three single-dimensional attributes of educational background, gender, and work fields are 226 784, 34 189, and 211, respectively. There are only three absolute fair cliques that satisfy 3-D attributes, which are as follows.

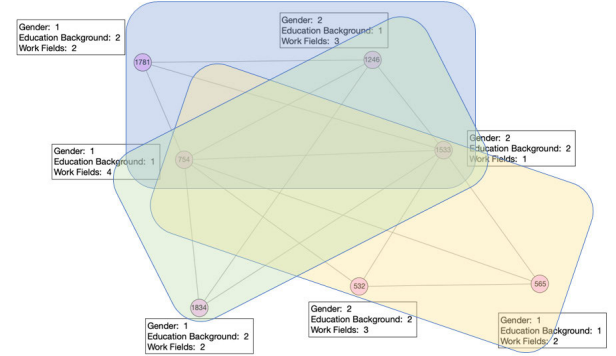


Fig. 9. Visualization of groups.

- 1) *Group 1*: {754, 1246, 1533, 1781}.
- 2) *Group 2*: {532, 565, 754, 1533}.
- 3) *Group 3*: {754, 1246, 1533, 1834}.

The visualization of these three absolute fair cliques is shown as Fig. 9. The number on the node is the user ID, and the label of each node indicate gender, educational background, and work fields of users. Fig. 10 is the visualization of Groups 1–3 in detail. Obviously, these three groups of users are fair in attributes of gender, educational background, and work fields.

The three absolute fair cliques of Fig. 10 contain some overlapping nodes, among them, nodes with IDs 754 and 1533 exist in all of Groups 1–3, which means that if either of these two users does not exist, there will be no absolute fair cliques in this 2000 users attribute social network. Therefore, we define a formula to calculate the overlapping degree of nodes in absolute fair cliques, and Lemma 2 gives the definition of the overlapping degree of nodes.

Lemma 2: Given a set of absolute fair cliques, the number of times each node v_i appears in the set of absolute fair cliques c_i , divided by the total number of absolute fair cliques N , is the overlapping degree $D_{overlap}$ of the node. The higher the $D_{overlap}$, the more important the node is in the network, and the greater the impact on the formation of absolute fair cliques

$$D_{overlap} = \frac{c_i}{N}, \quad (0 < D_{overlap} \leq 1). \quad (4)$$

In Fig. 9, the set of nodes covered by each rectangle is an absolute fair clique, where some nodes are covered by more than one rectangle, for example, ID 754 is a node covered by three rectangles in blue, yellow, and green, and ID 1246 is a node covered by two rectangles in blue and green. Importantly, the number of absolute fair cliques each node is covered is positively related to its overlapping degree $D_{overlap}$.

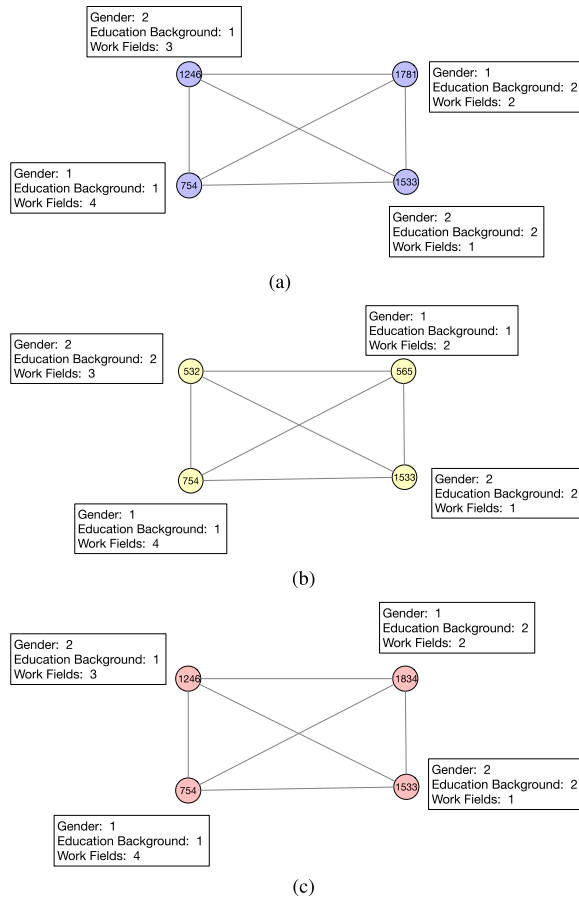


Fig. 10. Team formation strategy. (a) Group 1. (b) Group 2. (c) Group 3.

Table IX shows the overlapping degree of all nodes in this case study, where the nodes with IDs 754 and 1533 have the largest values of 1. That is to say, ID 754 is a key user for forming the absolute fair teams among the detected absolute fair cliques. Therefore, finding the users with the highest value of D_{overlap} can facilitate the detection of absolute fair cliques from attributed social networks.

VI. CONCLUSION

To achieve the responsible AI for social computing and further develop responsible computational social systems, this article investigated the problem of identifying absolute fair cliques from an attributed social network. Toward this end, we proposed an absolute fair clique model by incorporating fairness to cliques mining, and further proved it as a NP-hard problem. Due to the advantages of FCA for social network analysis, this article introduced a new concept called attributed equiconcept which can be used for characterizing the topological structure and attributes information of a cohesive subgraph. Then, a theorem and the corresponding algorithm *AFCMiner* were developed for extracting absolute fair cliques based on FCA from an attributed social network. We conducted extensive experiments using a large real-life social network in China, and the experimental results demonstrated the efficiency and effectiveness of *AFCMiner* algorithm. In addition, from the practical point of view, a case study was also presented to reveal the usefulness of the proposed absolute

fair clique model in an attributed social network. In the future, we plan to investigate the evolution of absolute fair cliques in dynamic attributed social networks with triadic FCA (TFCA) methodology for implementing the sustainable responsible computational social systems. Besides, weak fair clique model will also be explored by relaxing the fair constraints in the cliques.

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